REAL-TIME POSTURE
MONITORING SYSTEM

Using MediaPipe BlazePose & Computer Vision

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PROBLEM STATEMENT

Prolonged Sitting
Office workers sit 8–10 hrs/day with poor posture

Musculoskeletal Harm
Back pain, neck strain, spinal misalignment

No Real-Time Alerts
Existing tools lack live posture feedback

No Expert Needed
System must work without physiotherapist input

PROJECT OBJECTIVE

Build a real-time, webcam-based posture monitoring system that:

1
Detects posture quality continuously using a standard webcam

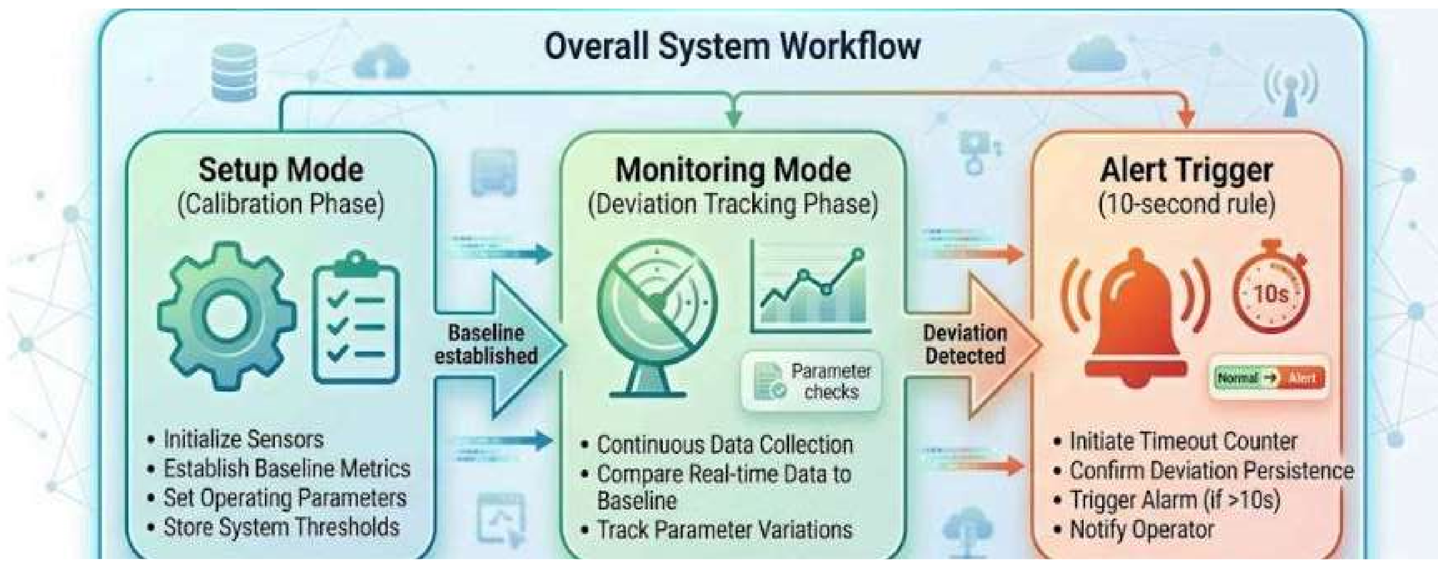
2
Personalizes thresholds via user calibration at session start

3
Alerts users after 10 seconds of sustained bad posture

4
Operates in real time at 28–32 FPS on standard hardware

5
Requires no physical sensors or wearable devices

SYSTEM ARCHITECTURE



Phase 1: Setup

- Initialize sensors
- Establish baseline metrics
- Set operating parameters
- Store system thresholds

Phase 2: Monitoring

- Continuous data collection
- Compare real-time data
- Track parameter variations
- Detect deviations

Phase 3: Alert

- Initiate timeout counter
- Confirm deviation persistence
- Trigger alarm (t > 10s)
- Notify operator

TECHNOLOGY STACK

🐍 Python 3.x

Core language

The primary programming environment for building the logic and integrating all libraries.

🦾 MediaPipe BlazePose

15 body landmarks detection

High-fidelity pose tracking used to identify skeletal key points in real-time.

📷 OpenCV (cv2)

Webcam capture & rendering

Handles video stream input, frame processing, and visual overlays on the output.

🔢 NumPy

Landmark math & deviation calc

Efficient array operations for Euclidean distance and coordinate calculations.

🔊 pyttsx3

Text-to-speech alerts

Cross-platform TTS library used for generating immediate audible posture warnings.

📊 Matplotlib

Performance visualization

Generates session reports and visualizes posture trends and FPS metrics.

INPUT & LANDMARK DETECTION

Live Webcam Feed

~30 FPS, adaptive resolution

MediaPipe BlazePose

Detects 15 skeletal landmarks per frame

Key Points Used

Shoulders, Hips, Ears (6 landmarks)

Confidence Threshold

Min detection & tracking: 0.5

Calibration Phase

User holds ideal posture → reference saved as NumPy arrays

Fig. 1 – Live Webcam with Pose Landmarks Overlay

ML METHODOLOGY

1. **Frame Capture**
   cv2.VideoCapture → flip → BGR to RGB

2. **Pose Detection**
   BlazePose extracts 15 landmarks per frame

3. **Landmark Extraction**
   Shoulders, Hips, Ears → NumPy arrays

4. **Deviation Scoring**
   Euclidean distance vs calibrated reference

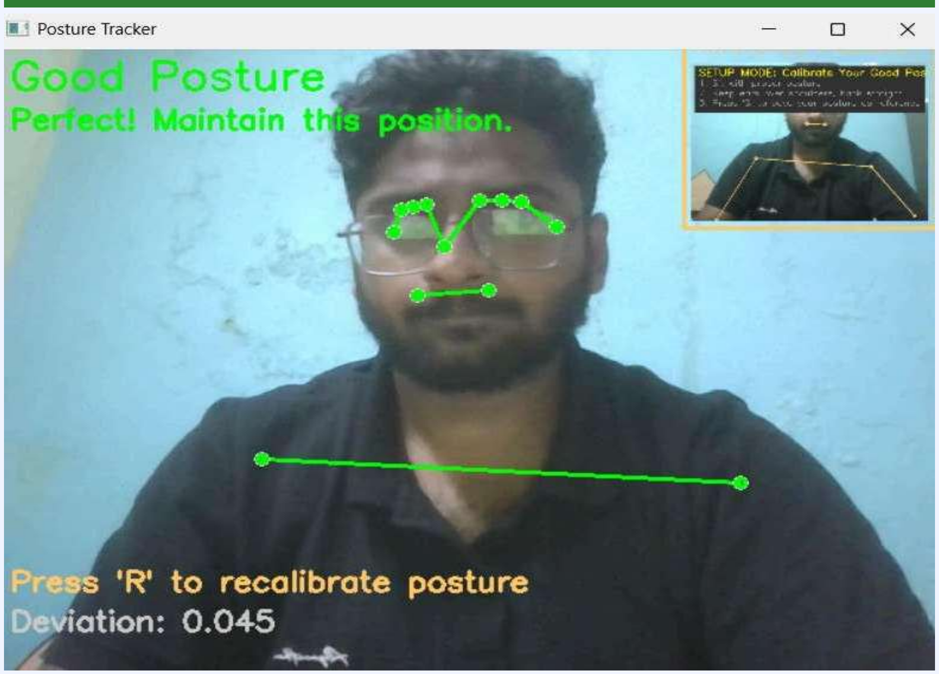
5. **Posture Classification**
   Good (<0.05) | Warning (0.05–0.10) | Bad (>0.10)

6. **Visual Feedback**
   Color-coded skeleton: Green / Orange / Red

7. **Alert Trigger**
   10-second rule → audio + banner alert

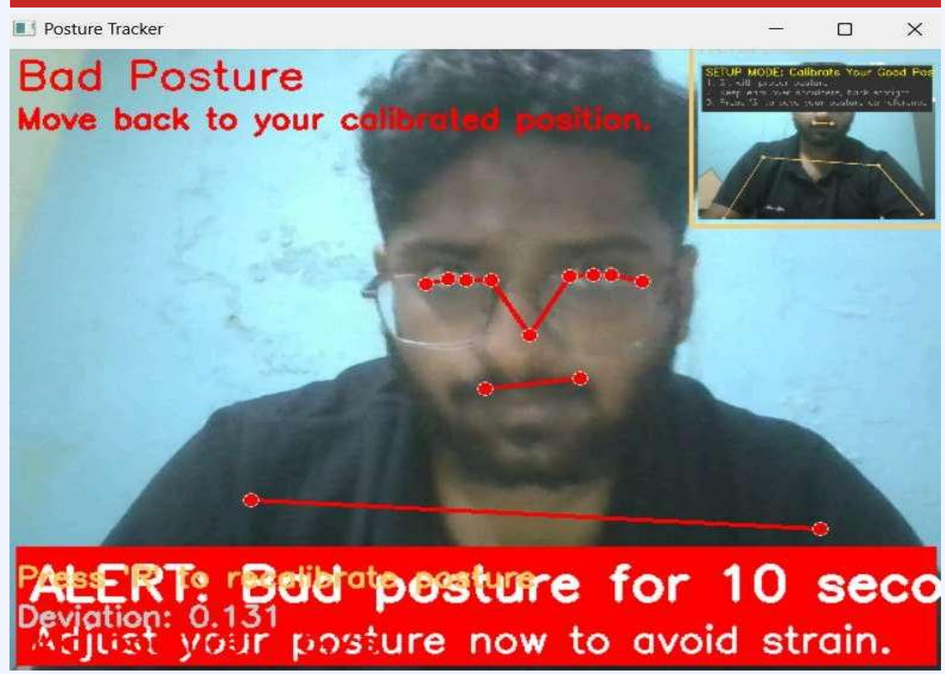
POSTURE DETECTION: GOOD vs BAD

✅ GOOD POSTURE (Deviation < 0.05)



Green skeleton overlay
Perfect! Maintain this position.

❌ BAD POSTURE (Deviation > 0.10)



Red skeleton + ALERT banner
10-second rule activated.

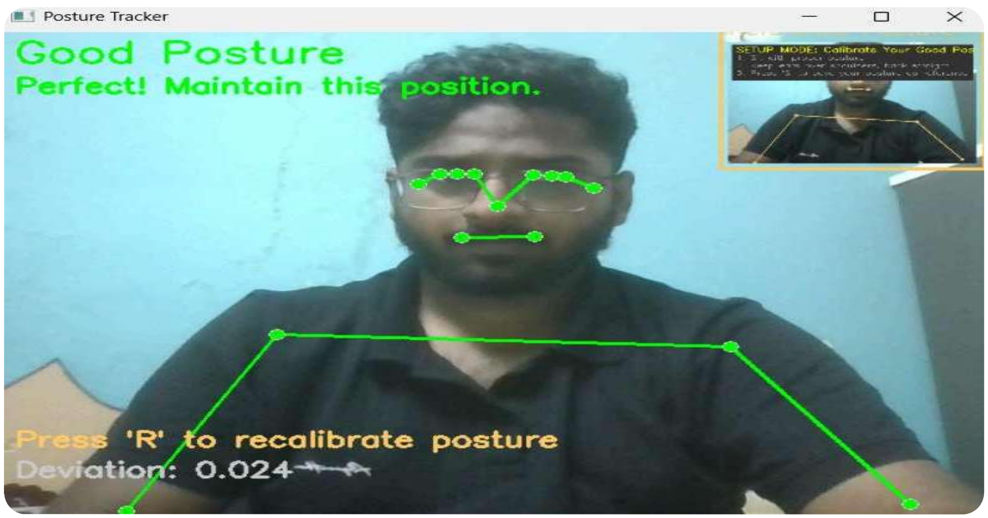
RESULTS & PERFORMANCE

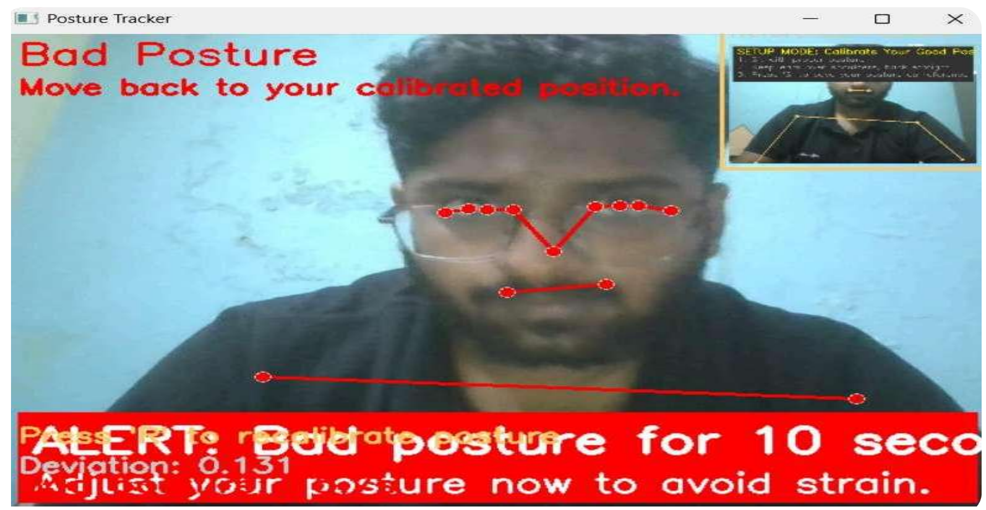
28–32
FPS
Real-Time Performance

0.5–2m
Range
Detection Distance

10s
Rule
Alert Responsiveness

Low
FPR
False Positive Rate





CONCLUSION & FUTURE SCOPE

Key Conclusions

- Achieved real-time posture feedback without any sensors
- Reference-based calibration adapts to any user body type
- 10-second alert rule minimizes false alarms effectively
- Runs at 28–32 FPS on standard laptop hardware
- Simple, cost-effective, scalable solution

Future Scope

- Multi-person posture tracking support
- Mobile deployment (Android / iOS app)
- ML classifier replacing deviation threshold
- Session history logging & trend analytics
- Integration with smart office IoT systems